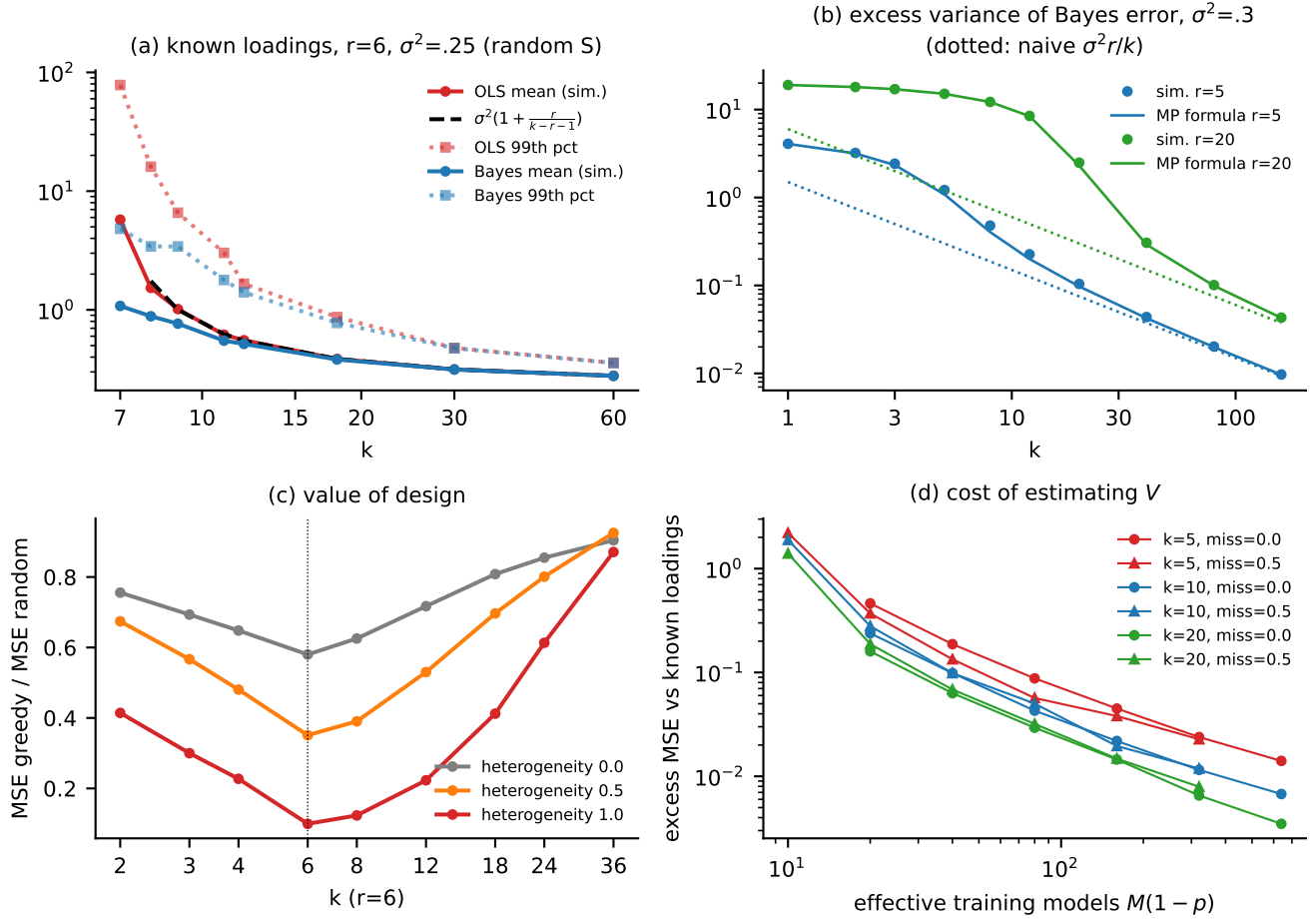
(a) known loadings, r=6, σ^2=.25 (random S)
OLS mean (sim.)
$\sigma^2(1+\frac{r}{k-r-1})$
OLS 99th pct
Bayes mean (sim.)
Bayes 99th pct
k
(b) excess variance of Bayes error, σ^2=.3
(dotted: naive $\sigma^2 r/k$)
sim. r=5
MP formula r=5
sim. r=20
MP formula r=20
k
(c) value of design
MSE greedy / MSE random
heterogeneity 0.0
heterogeneity 0.5
heterogeneity 1.0
k (r=6)
(d) cost of estimating V
excess MSE vs known loadings
k=5, miss=0.0
k=5, miss=0.5
k=10, miss=0.0
k=10, miss=0.5
k=20, miss=0.0
k=20, miss=0.5
effective training models $M(1-p)$